

# Question 12.118

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## 1 Question:

The eigenvalues of the matrix,  $\mathbf{A}^{-1}$ , where  $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$  are

a) 1 and 1/2

c) 2 and 3

b) 1 and 1/3

d) 1/2 and 1/3

## 2 Solution:

The eigenvalues of a matrix  $\mathbf{A}$  are given by the roots of the characteristic polynomial, which is defined as:

$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

where  $\lambda$  represents the eigenvalues and  $\mathbf{I}$  is the identity matrix of the same size as  $\mathbf{A}$  ( $\mathbf{I}_2$ ).

To find the eigenvalues of  $\mathbf{A}^{-1}$ , we can use the property that the eigenvalues of the inverse of a matrix are the reciprocals of the eigenvalues of the original matrix. This can be proven using the fact

$$\begin{aligned} \mathbf{A}\mathbf{v} &= \lambda\mathbf{v} \implies \mathbf{A}^{-1}\mathbf{A}\mathbf{v} = \mathbf{A}^{-1}(\lambda\mathbf{v}) \\ &\implies \mathbf{v} = \lambda\mathbf{A}^{-1}\mathbf{v} \\ &\implies \mathbf{A}^{-1}\mathbf{v} = \left(\frac{1}{\lambda}\right)\mathbf{v} \end{aligned}$$

Therefore, we first need to find the eigenvalues of  $\mathbf{A}$ .

The characteristic polynomial of  $\mathbf{A}$  is given by:

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \det \begin{vmatrix} 2 - \lambda & 1 \\ 0 & 3 - \lambda \end{vmatrix} = (2 - \lambda)(3 - \lambda) - 0 = (2 - \lambda)(3 - \lambda)$$

Setting this equal to zero gives us the eigenvalues of  $\mathbf{A}$ :

$$(2 - \lambda)(3 - \lambda) = 0$$

Thus, the eigenvalues of  $\mathbf{A}$  are  $\lambda_1 = 2$  and  $\lambda_2 = 3$ .

Now, the eigenvalues of  $\mathbf{A}^{-1}$  are the reciprocals of these eigenvalues:

$$\mu_1 = \frac{1}{\lambda_1} = \frac{1}{2}, \quad \mu_2 = \frac{1}{\lambda_2} = \frac{1}{3}$$

Therefore, the eigenvalues of  $\mathbf{A}^{-1}$  are  $\frac{1}{2}$  and  $\frac{1}{3}$ . The correct answer is option (d):  $1/2$  and  $1/3$ .